When Predictive Surrogates Fail as RL Environments: A Calibrated Physical Twin as Soft Regularizer for HVAC Control*

Almaz Sapargali[a,*]

[a] *Al-Farabi Kazakh National University, Faculty of Information Technology and Artificial Intelligence, 71 al-Farabi Avenue, Almaty, 050040, Kazakhstan*

Abstract

Reinforcement learning controllers for heating, ventilation and air conditioning (HVAC) systems are typically trained on neural-network surrogates because high-fidelity simulators are too slow for the millions of environment steps that modern policy-gradient methods consume. A natural assumption is that a more physically faithful surrogate produces a better training environment. We test that assumption on the BOPTEST `bestest_air` testcase and report a clear negative result: a calibrated physical twin with explicit zone thermal capacitance achieves a 24-hour rollout root-mean-square temperature error of 0.64 °C (versus 1.47 °C uncalibrated), yet collapses as a stand-alone reinforcement learning training environment, producing a live closed-loop RMSE above 4 °C. We resolve the gap by repurposing the calibrated twin as a frozen *soft physical regularizer* for a smoother data-driven surrogate. The resulting hybrid backend sustains 1787 environment steps per second on a single CPU thread, corresponding to an 85.0× **speed-up** over the standard BOPTEST RTE HTTP–Docker deployment used in production benchmarking, under the same 15 min control protocol, while restoring live closed-loop RMSE below 0.65 °C. The optimal regularization strength is controller-family specific: $\lambda_{\text{temp}} = 0.10$ for thermostatic proximal-policy optimization, but $\lambda_{\text{temp}} = 0.00$ for both hierarchical reinforcement learning and multi-objective reinforcement learning with a 17-dimensional observation interface. The full pipeline follows the surrogate-development protocol of Hou and Evins (2024), with all numerical justifications provided in Supplementary Tables S1–S11.

Keywords: HVAC control, deep reinforcement learning, digital twin, physics-informed machine learning, BOPTEST, multi-objective reinforcement learning

1. Introduction

Heating, ventilation and air conditioning (HVAC) systems account for the largest share of energy use in buildings, and buildings in turn account for roughly a third of global final energy demand (Pérez-Lombard et al., 2008; International Energy Agency, 2024). Decarbonising this end use depends as much on better operation of existing systems as on better hardware: advanced control strategies that exploit weather forecasts, internal-gain signals, and slow envelope dynamics have repeatedly been shown to deliver double-digit energy savings without compromising occupant comfort (Drgoňa et al., 2020; Mason and Grijalva, 2019). Among these strategies, deep reinforcement learning (DRL) has attracted particular attention because it learns a closed-loop policy directly from interaction with a building model, in principle without requiring hand-tuned MPC weights or expensive solver invocations at run time.

A practical obstacle stands between the promise and the deployments, however. Modern policy-gradient algorithms such as PPO consume on the order of 10^6–10^7 environment steps before they converge (Schulman et al., 2017; Henderson et al., 2018), and high-fidelity building simulators are far too slow to provide those steps. The BOPTEST testbed (Blum et al., 2021; Arroyo et al., 2022) addresses reproducibility by exposing a standardised HTTP API on top of the IBPSA Modelica Buildings library (Wetter et al., 2014), but on commodity hardware that API delivers fewer than two-dozen environment steps per second — two orders of magnitude below what a single PPO training run requires per CPU thread. The standard work-around in the literature is to replace the simulator with a neural-network surrogate trained from logged trajectories (Zhang et al., 2019; Amasyali and El-Gohary, 2018; Kazmi et al., 2018).

When practitioners adopt the surrogate work-around they implicitly adopt a stronger assumption as well: that a more physically faithful surrogate must be a better RL training

---

*Reproducibility: full code, configuration, trained models, and all appendix tables (S1–S11) are available at `<repository URL placeholder>`.

*Corresponding author.
Email address: almaz.sapargali2001@gmail.com (Almaz Sapargali)

environment. The assumption is intuitive, because a faithful surrogate keeps the policy's experience close to what it would have been under the real simulator. It is also under-tested in the literature. Most surrogate-vs.-RL studies report either predictive validity (one-step or multi-horizon root-mean-square error against held-out trajectories) or closed-loop controller performance, not the relationship between the two (Hou and Evins, 2024; Coraci et al., 2025). In Section 5 we test the assumption explicitly on BOPTEST `bestest_air` and report a clear negative result.

The resolution we pursue keeps the calibrated physical twin in the pipeline but moves it out of the rollout. The smoother data-driven surrogate remains the training environment; the physical twin is frozen after Stage A/B/C calibration and enters the policy training loss only as a soft physical regulariser, penalising actions on which the two surrogates physically disagree. This hybrid construction is fast (Section 4.6) and faithful (Section 6), and it exposes a non-trivial empirical structure: the optimal regularisation strength is *controller-family specific* rather than universal.

Contributions

This paper makes the following contributions:

1. A physics-informed surrogate (`v3.5`) of the BOPTEST `bestest_air` testcase, calibrated through a three-stage inverse procedure (Stage A telemetry preprocessing, Stage B explicit identification of zone thermal capacitance C_{zon}, Stage C residual head calibration), achieving a 38% reduction in one-step temperature RMSE and a 40% reduction in power MAE over the uncalibrated baseline.
2. A clear empirical demonstration that predictive validity does not transfer to reinforcement learning training utility: the calibrated twin achieves a 24-hour rollout RMSE of 0.64 °C on held-out trajectories yet drives a closed-loop BOPTEST RMSE above 4 °C when used as a stand-alone RL training environment.
3. A hybrid surrogate construction (`hybrid_l010`) in which the calibrated twin acts as a frozen *soft physical regularizer* on a smoother control-oriented surrogate, restoring closed-loop transfer RMSE below 0.65 °C on both `peak_heat_window` and `typical_heat_window`

scenarios while sustaining 1787 environment steps per second on a single CPU thread (85.0× speed-up over the live BOPTEST RTE HTTP loop at the same 15 min control protocol; see Table 1).

4. A controller-family-specific finding: the optimal physical regularization strength differs across controller architectures ($\lambda_{\text{temp}} = 0.10$ for thermostatic PPO, $\lambda_{\text{temp}} = 0.00$ for HDRL and 17-D MORL), with an interpretation grounded in observation-space geometry and control hierarchy.
5. A pre-registered transferability characterisation across three additional BOPTEST hydronic testcases that decomposes the recipe into a surrogate component, which transfers structurally (identified C_{zon} ratio 1.91 ± 0.03 on $N = 3$), and a controller component, which fails on all three under a frozen direct-supply-temperature policy. The decomposition is the central Block 3 result and is reported in Section 7.
6. A reproducible BOPTEST-based evaluation protocol, with full Hou and Evins (2024)-compliant numerical justification for sample generation, preprocessing, feature significance and independence, scaling, hyperparameters, architecture, and replicative-plus-predictive validity, provided as Supplementary Tables S1–S11.

The remainder of the paper is organised as follows. Section 2 positions the work against three literatures (DRL for HVAC, surrogate modelling for buildings, and physics-informed machine learning). Section 3 describes the two surrogates, the hybrid loss, and the controller families. Section 4 fixes the evaluation protocol, introduces the Hou and Evins (2024) compliance and the pre-registration audit chain. Sections 5–7 report the three blocks of results in order: surrogate fidelity, controller performance, and transferability across testcases. Section 8 explains the mechanisms behind the controller-family-specific findings and states the threats to validity, and Section 9 closes with a tight scope statement and the natural follow-up.

2. Related Work

2.1. Deep reinforcement learning for HVAC control

Deep reinforcement learning has been applied to HVAC supervisory control for the better part of a decade, starting with discrete-action Q-learning controllers (Wei et al., 2017; Mnih et al., 2015) and moving rapidly to continuous-action policy-gradient methods such as PPO and SAC (Schulman et al., 2017; Haarnoja et al., 2018). Recent reviews catalogue dozens of such studies on platforms ranging from in-house EnergyPlus wrappers through Sinergym (Jiménez-Raboso et al., 2021) to the BOPTEST suite (Blum et al., 2021); surveys consistently report that the bottleneck to deployment is not algorithmic but environmental and statistical (Mason and Grijalva, 2019; Al Sayed et al., 2024). Two recurring weaknesses are particularly relevant here. First, the vast majority of DRL-HVAC studies report a single seed or at most three, making energy-comfort comparisons fragile under environment stochasticity (Henderson et al., 2018; Agarwal et al., 2021; Hedayat et al., 2025). Second, the choice of *which* surrogate to train on is rarely subjected to scrutiny: most studies train and evaluate on the same surrogate without an out-of-loop fidelity check, or evaluate on a different simulator without a clear bridge from training-surrogate accuracy to deployment performance (Zhang et al., 2019; Chen et al., 2019). The *controller-side gap* the present paper addresses is the second: existing DRL studies rarely separate the question of surrogate fidelity from the question of controller performance.

Hierarchical decomposition has been proposed as one structural response to long-horizon HVAC control problems, splitting setpoint planning from low-level tracking (Bacon et al., 2017; Nachum et al., 2018; Liao et al., 2025). Multi-objective formulations (Roijers et al., 2013; Yang et al., 2019; Hayes et al., 2022) are similarly motivated by the natural energy-comfort trade-off in HVAC and have been used to expose tunable Pareto fronts at deployment time (Gao et al., 2024; Sun et al., 2024). We adopt both families as evaluation arms in Section 6.

2.2. Surrogate and digital twin models for building energy

Surrogate modelling for building energy is a mature subfield. Reviews of data-driven approaches list dozens of architectures with reported one-step and short-horizon RMSE numbers on benchmark testcases (Amasyali and El-Gohary, 2018). Recent contributions have moved from pure black-box regression toward physically informed surrogates that retain explicit thermodynamic parameters: low-order resistance–capacitance (RC) models (Bacher and Madsen, 2011; Berthou et al., 2014; Bünning et al., 2020) or RC–neural-ODE hybrids (Chen et al., 2018, 2022) calibrated by inverse procedures (Coraci et al., 2025; Wang et al., 2025). The recent protocol of Hou and Evins (2024) consolidates these efforts into a four-stage development pipeline with explicit Reporting-Level-3 disclosure requirements; we adopt it as the methodological standard for Block 1 of the present work.

A common feature of surrogate-fidelity studies, however, is that they evaluate accuracy along trajectories that always start from a real state. That setting is informative about *predictive* accuracy but only indirectly about the surrogate's value as a closed-loop RL training environment, where every rollout is bootstrapped from the surrogate's own output. Picard et al. (2017) highlights an analogous distinction in the MPC context, and prior reviews of surrogate-based MPC and RL emphasise the same point in passing (Mason and Grijalva, 2019; Drgoňa et al., 2020), but the failure mode is rarely quantified directly. The *fidelity-side gap* addressed here is precisely that: showing the well-defined regime in which a better predictive surrogate is not a better RL training environment.

2.3. Physics-informed and physics-guided machine learning

The physics-informed machine-learning literature spans a wide range of constructions, from hard-constraint physics-informed neural networks (PINNs) (Raissi et al., 2019) through theory-guided data science (Karpatne et al., 2017; Willard et al., 2022) to neural ordinary differential equations (Chen et al., 2018). Our construction is deliberately on the soft-regularisation side: the physical model is not enforced as a hard constraint, and it is not even part of the policy's forward pass. It enters only as a penalty term that the policy can violate at a cost. This is closer in spirit to auxiliary tasks (Jaderberg et al., 2017) and Dyna-style world models (Sutton, 1991; Hafner et al., 2025) than to PINNs, and we position it as a controller-side regularisation strategy rather than as a surrogate-architecture contribution.

2.4. BOPTEST benchmarking

We adopt the BOPTEST benchmark (Blum et al., 2021; Arroyo et al., 2022; IBPSA Project 1, 2024) as the reproducible testbed throughout this study. Its `bestest_air` testcase exposes a single-zone air-side HVAC system with a documented comfort band, a TMY weather file, and a composite safety/energy KPI (m_s) that weights duration and severity of comfort violations against energy use. Built-in PI control is available as a default reference; we use it as the normaliser for Block 2 results (Section 6.1). Block 3 extends the evaluation to three additional BOPTEST hydronic testcases under a pre-registered manifest.

2.5. Position of this work

This work sits at the intersection of the three gaps above. It isolates the surrogate-fidelity question from the controller-performance question through an explicit two-stage protocol (Block 1 then Block 2); it shows the well-defined failure mode of fidelity-driven surrogates as RL training environments (Section 5.4); and it introduces hybrid soft regularisation as the resolution, evaluated across three controller families. Block 3 adds a pre-registered transferability characterisation across three hydronic BOPTEST testcases and reports the recipe's decomposable behaviour: the surrogate component transfers structurally and the controller component does not, under the chosen frozen-controller scope (Section 7).

3. Methodology

The methodology is organized in four blocks: (i) the control-oriented data-driven surrogate `v3`, (ii) the physically informed surrogate `v3.5` together with its three-stage inverse calibration pipeline, (iii) the hybrid backend that combines the two, and (iv) the controller families and benchmark protocol. Subsection 3.5 summarizes the information flow between the components.

3.1. Control-oriented surrogate `v3`

The control-oriented surrogate `v3` is a two-headed feed-forward neural network. The temperature head predicts the next-step zone temperature $T_{\mathrm{zone},t+1}$ and the power head predicts the total HVAC power P_{total} at the same step, both as functions of a smooth control-oriented observation stack derived from direct-supply-temperature trajectories collected on BOPTEST `bestest_air`. Both heads share three hidden layers of width 128 with ReLU activations; outputs are unscaled by the standardisation parameters listed in Supplementary Table S7. The model is trained to minimise the mean-squared error against the held-out prepared 15-minute corpus described in Supplementary Tables S1–S2; the full training hyperparameters (Adam optimiser, learning rate 3×10^{-4}, batch 256, early stopping with patience 20 on validation loss) are tabulated in Supplementary Table S8. The role of `v3` in this paper is deliberately functional rather than physical: its gradient surface is smooth enough that PPO clip-ratio updates remain stable through 10^6 environment steps, and its inference cost is small enough to support the throughput reported in Section 4.6.

3.2. Physically informed surrogate `v3.5` and inverse calibration

The physically informed surrogate `v3.5` is a RC–neural-ODE construction. The zone temperature evolves according to a continuous balance $\mathrm{d}T_{\text{zone}}/\mathrm{d}t = \big(q_{\text{net}}(s,a) - Q_{\text{wall}}(s)\big)/C_{\text{zon}}$ where C_{zon} is the lumped zone thermal capacitance (a single learnable scalar identified during Stage B), Q_{wall} is an envelope-loss term parameterised by the testcase weather and an UA coefficient, and $q_{\text{net}}(s,a)$ is a residual heat-flow network with two hidden layers of width 64 that absorbs non-RC physics (solar gains through windows, internal gains, residual nonlinearity in UA). The total HVAC power is predicted by a separate head $P_{\text{total}}(s,a)$ trained jointly. Integration uses a fixed-step fourth-order Runge–Kutta scheme with a one-minute sub-step over the 15-minute control interval. Inverse calibration proceeds in three stages.

Stage A — Telemetry preprocessing.. Five preprocessing operations are applied in sequence to the raw BOPTEST telemetry: (1) latency compensation of the actuator-to-zone delay, (2) temperature bias removal against a low-pass reference, (3) affine normalisation of the power channel from raw kW into the range observed under direct-supply-temperature control, (4) rolling denoise of high-frequency measurement noise with a window matched to the sensor time constant, and (5) causal recomputation of the ΔT_{zone} target so that the prediction is strictly one-step-ahead. Each operation has a numerical selection criterion documented in Supplementary Table S3, and each is reversible at inference time through the parameters recorded in Supplementary Table S7.

Stage B — Identification of C_{zon}.. Stage B subselects an excitation-rich window of the prepared corpus using the top-5% quantile of $|\Delta T_{\mathrm{zone}}|$ and splits it 80/20 into train and validation while keeping every episode atomic to avoid trajectory leakage. With the residual heads frozen at their Stage A initialisation, C_{zon} is optimised against 24-hour rollout mean-squared error using L-BFGS for at most 120 epochs. The result on `bestest_air` is $C_{\mathrm{zon}} = 4.413 \times 10^5\,\mathrm{J\,K^{-1}}$ (Section 5.2).

Stage C — Residual head calibration.. With C_{zon} frozen at its Stage B value, the residual heat-flow network q_{net} and the power head P_{total} are updated on the full prepared corpus to absorb the remaining non-RC physics. The residuals are centred and tightened relative to the raw model (Supplementary Table S11; Figure 2b in the results), and the resulting `v3.5` checkpoint is the canonical anchor used throughout the rest of the paper.

3.3. Hybrid backend: `v3` dynamics with `v3.5` as soft regularizer

The hybrid backend treats the calibrated `v3.5` model as a frozen *physical censor.* Reinforcement learning rollouts evolve under the smoother `v3` dynamics, and the policy is penalized at training time for actions on which the two surrogates physically disagree:

$$\mathcal{L}_{\mathrm{total}} = \mathcal{L}_{\mathrm{PPO}}\Big(\pi_\theta;\ \mathtt{v3}\Big) + \lambda_{\mathrm{temp}} \left\| T_{\mathtt{v3}}(s,a) - T_{\mathtt{v3.5}}(s,a) \right\|_2^2 + \lambda_{\mathrm{power}} \left\| P_{\mathtt{v3}}(s,a) - P_{\mathtt{v3.5}}(s,a) \right\|_2^2. \quad (1)$$

Here λ_{temp} and λ_{power} are the temperature and power disagreement weights, the only two hyperparameters of the hybrid construction. Crucially, `v3.5` never enters the policy's forward pass; it only enters the loss. This decoupling is what allows the hybrid backend to retain the smooth control-oriented training surface of `v3` while inheriting the physical sanity of `v3.5`.

The intuition for why this beats both endpoints is straightforward. Pure `v3` training optimises a control-oriented surrogate that has no physical anchor: the policy can drift into regions where the surrogate is inaccurate without paying any penalty, and the resulting mis-specification only becomes visible on live BOPTEST. Pure `v3.5` training is faithful but, as shown in Section 5, structurally unsuited to closed-loop training because the policy learns to compensate for residual integration errors as if they were real disturbances. The hybrid backend lets PPO walk along the smoother `v3` surface while pulling it back toward `v3.5`-consistent actions whenever the two disagree, restoring closed-loop transfer to within 0.65 °C on both peak and typical heating windows (Section 6).

3.4. Controller families and benchmark protocol

Four controllers are evaluated under an identical BOPTEST evaluation protocol (Section 4.4).

- *PI baseline*: the BOPTEST built-in proportional-integral controller, invoked by passing an empty action dictionary to the standard BOPTEST step API. Used as the normalising reference for all RL results (Section 6.1).

- *Thermostatic PPO*: single-level PPO on the `no_delta_t_powerlog_tzone` observation stack. Actor and critic are two-layer MLPs of width 64 with tanh activations. Continuous action: supply temperature in $[20, 60]$ °C.

- *HDRL*: a hierarchical decomposition with a high-level setpoint planner that updates every hour and a low-level supply temperature tracker that updates every 15 minutes. The low-level policy is pre-trained for 2×10^5 steps with a fixed high-level setpoint before joint fine-tuning.

- *MORL*: preference-conditioned PPO with the 17-D TSup-style observation stack and a comfort/energy preference vector $w \in \Delta^2$ supplied as an additional input. Training samples w uniformly from the simplex; inference uses a fixed w.

Implementation details and training-time hyperparameters for each controller family are tabulated in Supplementary Table S8.

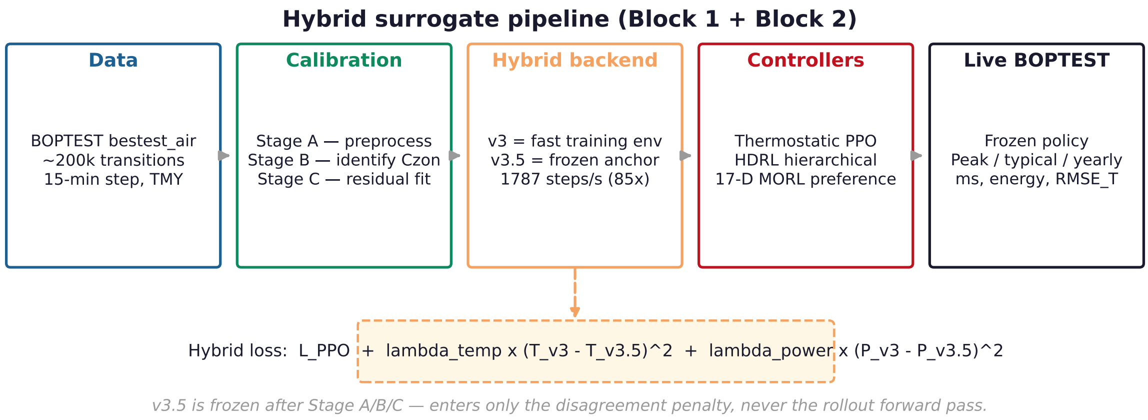

Figure 1: Information flow in the hybrid surrogate backend. The frozen physically informed model `v3.5` enters the policy training loss through the disagreement terms in Eq. (1) but not the policy's forward pass; the rollout environment is the smoother control-oriented surrogate `v3`.

3.5. Information flow

Figure 1 schematises the information flow. `v3` forms the rollout environment for PPO; `v3.5` is frozen and contributes only the disagreement penalty in Eq. (1). The same trained policy is evaluated on live BOPTEST without any further fine-tuning; the evaluation protocol in Section 4.4 is fixed across all controller families and all results sections.

4. Experimental Setup

4.1. Testbed and serving layer

All experiments use the BOPTEST `bestest_air` testcase, a single-zone air-side HVAC reference building. BOPTEST is served through the `boptest_rte` container, which exposes the standard BOPTEST HTTP API to both surrogate training and live closed-loop validation. The `bestest_air` envelope is a single thermal zone ($48\,\mathrm{m}^2$ floor area, lightweight construction following the ANSI/ASHRAE 140 BESTEST 600 baseline), heated by an idealised air-side unit that accepts a supply-temperature command in $[20, 60]\,°\mathrm{C}$. Weather is drawn from the BOPTEST default TMY file for Brussels, Belgium, with a 15-minute control interval and a comfort band of $[21, 24]\,°\mathrm{C}$ throughout the heating season. The Block 3 hydronic testcases share this single-zone envelope class but differ in actuator interface; the testcase manifest is summarised in Table 2 of Section 5.1 and detailed in Section 7.

4.2. Datasets and sample generation

Three transition corpora support the experiments. The `v3` hourly corpus is the original logging frequency of the direct-supply-temperature trajectories and is used only to bootstrap the data-driven surrogate; the `v3.5` prepared 15-minute corpus is the canonical training set after the Stage A preprocessing of Section 3.2; and the `v3.5` collected 15-minute corpus is the raw form of the same trajectories before preprocessing, retained for ablation. Supplementary Table S1 lists the provenance, parameter ranges, and physical rationale of each corpus, and Supplementary Table S2 documents the numerical justification of the chosen sample sizes (including the learning-curve check that fixes the canonical corpus at $\sim 2\times 10^5$ transitions). The canonical artefact for the calibrated twin is the prepared 15-minute corpus; the hourly corpus is used only for the `v3` pre-training run referenced in Section 3.1.

4.3. Preprocessing, scaling and feature engineering

Preprocessing follows the Stage A pipeline documented in Supplementary Table S3 (latency compensation, bias removal, power affine normalisation, rolling denoise, causal ΔT recomputation). Per-channel scaling parameters and the reverse transform used at inference time are listed in Supplementary Table S7. Pairwise input independence is verified in Supplementary Table S5 through Pearson correlation and mutual-information checks; no input pair exceeds the documented thresholds.

4.4. Evaluation Protocol

The evaluation protocol is fixed across all controller families and **never varies** between Section 5 and Section 6.

Scenarios.. Two live BOPTEST closed-loop windows are used as the primary transfer benchmarks:

- `peak_heat_window`: simulated start $t_{\text{start}} = 14$ days after 1 January, duration 14 days, capturing the coldest contiguous window of the Brussels TMY file.
- `typical_heat_window`: simulated start $t_{\text{start}} = 108$ days after 1 January, duration 14 days, capturing a milder average heating period.

Yearly validation uses the full BOPTEST year with KPI windows defined by the BOPTEST specification.

Simulation step and comfort band.. The simulation step is fixed at $\Delta t = 900\,\text{s}$ (15 min). The comfort band is $[T_{\text{low}}, T_{\text{high}}] = [21\,°\text{C}, 24\,°\text{C}]$.

Metrics..

- m_s: the BOPTEST composite KPI combining thermal discomfort and energy use; lower is better.
- `violation %`: fraction of simulation steps spent outside the comfort band.
- `energy`: total final energy consumption in kWh.
- `RMSE_T`: root-mean-square error of zone temperature against a reference trajectory; reported separately for one-step replicative validity and for multi-horizon predictive rollout.
- `first_divergence_step`: the first simulation step at which the surrogate-trained policy's trajectory diverges from the live BOPTEST trajectory by more than a fixed threshold (transfer-realism diagnostic).

Statistical reporting..

- Predictive validity numbers (Section 5) report bootstrap 95% confidence intervals over rollout windows, inherited directly from `outputs/surrogate_v35_rollout_prepared_15min_powe`

- Canonical controller results (Section 6) are reported as mean $\pm$ std over $N = 3$ random seeds (`42`, `43`, `44`) for the thermostatic hybrid, the pure `v3` baseline, and the MORL canonical configuration. HDRL is reported single-seed; this is acknowledged as a limitation in Section 8.

4.5. Reproducibility statement

All code, configuration, trained models, and the eleven supplementary tables are released through the project repository and the associated archival deposit; the URL and DOI are listed on the manuscript metadata page. The pre-registered audit chain comprises four git-anchored commits: (a) MORL canonical-selection pre-registration (`93df9b3`); (b) MORL post-N=5 canonical falsification (`62dc859`); (c) Block 3 transferability pre-registration (commit recorded in the manifest field `audit.pre_registration_commit_sha` of `configs/block3_testcase_manifest.yaml`); and (d) Block 3 closure (`audit.block3_close_commit_sh` The BOPTEST version is `0.7.0`; the testcase identifiers are `bestest_air`, `bestest_hydronic_heat_pump`, `bestest_hydronic`, and `singlezone_commercial_hydronic`. The software stack is Python 3.11, PyTorch 2.1, NumPy 1.26, and Stable-Baselines3 2.2 (Raffin et al., 2021). All experiments run on a single CPU thread (Intel Core i7-class, 32 GB RAM, no GPU required); random seeds are fixed in `configs/seeds.yaml`. Numerical values in Tables 2–4 are reproduced automatically from the corresponding CSVs in `reports/` by `paper/build_paper_tables.py`, so every cell in the main tables is traceable to a deterministic source file.

4.6. Runtime characteristics

A throughput comparison between the live BOPTEST RTE HTTP loop and the three surrogate backends is reported in Table 1. All backends were exercised at the same 15 min control protocol on a single CPU thread, with 100 episodes of 96 transitions each (9,600 total transitions per backend). Surrogate timings exclude one-shot model loading.

The live BOPTEST RTE HTTP backend reaches 21.0 environment steps per second — the same path that downstream RL training and live closed-loop validation use, including HTTP serialization overhead, Flask request handling, JSON encoding, and Docker container isolation. The control-oriented `v3` surrogate reaches 4626.4 steps/s (220.2× speed-up), the calibrated physical twin `v3.5` reaches 2399.5 steps/s (114.2×), and the canonical hybrid backend reaches 1786.8 steps/s (85.0×). The hybrid is slower than `v3` alone because it evaluates both networks plus the disagreement terms in Eq. (1) on every step; 85× is the practical speed-up factor relevant for the rest of the paper, since all canonical RL training is done on the hybrid backend.

Scope of the comparison.. The 85× figure compares the surrogate to the *standard BOPTEST RTE deployment* — the HTTP–Docker stack used by the BOPTEST community for production benchmarking and used here for all downstream live closed-loop validation. This comparison conflates two costs: the intrinsic Modelica/FMU simulation cost and the deployment overhead (HTTP roundtrip, JSON serialization, Flask request handling, container isolation). To isolate the simulation cost, we additionally implemented a benchmark of direct FMU loading via the `fmpy` co-simulation interface (script `evaluation/build_speed_benchmark_fmu_direct.py`). The BOPTEST `bestest_air` FMU (`boptest_rte/testcases/bestest_air/models/wrapped.fmu`) ships with a Linux `x86_64` binary only (`binaries/linux64/wrapped.so`); the archive contains no Windows or macOS native binary, which is the reason BOPTEST is conventionally deployed in a Linux container. Because all measurements in this study were collected on a Windows host (matching the host of the live HTTP benchmark for cross-comparability), the direct FMU benchmark cannot be run in-process on the same machine without a Linux runtime. The script is committed with explicit reproducibility instructions for WSL2 Ubuntu and Docker, and the platform constraint is logged at `reports/speed_benchmark_fmu_direct_platform_log.txt`.

We accordingly frame the reported speed-up as a comparison against the *practical BOPTEST deployment configuration* rather than against bare FMU evaluation. A direct FMU–surrogate comparison remains a deferred reproducibility item: when executed on a Linux host, it will produce a third row in Table 1 that isolates the deployment-overhead component of the current 85× figure. We do not anticipate the surrogate losing its speed advantage, because the surrogate forward pass is a fixed-cost tensor evaluation (∼0.5 ms

Table 1: Throughput comparison on CPU at the same 15 min control protocol (100 episodes × 96 steps = 9,600 transitions per backend). The live BOPTEST RTE row uses the same HTTP API path that downstream RL training and live closed-loop validation actually use, so the speed-up factor in the last column reflects the practical, not the idealized, ratio. Surrogate timings exclude one-shot model loading. Source: `reports/speed_benchmark_table.csv`.

Backend	Steps/s	Median step (ms)	P95 step (ms)	Speed-up
BOPTEST RTE (HTTP)	21.0	14.915	19.251	1.0×
`v3` surrogate	4,626.3	0.165	0.273	220.2×
`v3.5` calibrated	2,399.5	0.359	0.495	114.2×
`hybrid_1010`	1,786.8	0.507	0.648	85.0×

on CPU) whereas the FMU integrates a constrained nonlinear Modelica system per step; however, the precise ratio against bare FMU is not currently reported and is acknowledged as a limitation in Section 8.

5. Results I: Digital Twin Fidelity

This section answers the question *"How accurate is the surrogate as a model of the building?"* It deliberately does *not* answer "Does it train good RL controllers?" — that question is addressed in Section 6.

5.1. Architecture comparison

Table 2 compares the three surrogate variants used throughout this paper. The data-driven `v3` model has no identifiable physical parameter; the calibrated `v3.5` model exposes C_{zon} explicitly through Stage B; and the canonical `hybrid_1010` backend uses `v3.5` only as a regulariser ($\lambda_{\mathrm{temp}} = 0.10$, $\lambda_{\mathrm{power}} = 5 \times 10^{-5}$) without admitting it into the policy's forward pass. The same table also previews the closed-loop transfer numbers that will be developed in Section 6: pure `v3` and `hybrid_1010` both reach sub-degree live RMSE on the peak and typical heating windows, while pure `v3.5` exceeds $4\,°\mathrm{C}$ on both — the divergence between predictive accuracy and RL training utility that motivates the rest of the paper.

Table 2: Architecture comparison of the three surrogate variants. `v3` is a data-driven feed-forward model on direct-TSup trajectories; `v35_calibrated` is the physically informed RC-NeuralODE after Stage A/B/C calibration with explicit zone thermal capacitance C_{zon}; `hybrid_1010` is the canonical hybrid backend ($\lambda_{\text{temp}} = 0.10$, $\lambda_{\text{power}} = 5 \times 10^{-5}$). Block 1 fidelity columns are reproduced from `reports/block1_surrogate_final_metrics.csv`; live-transfer and divergence columns from `outputs/block13_*`.

Variant	Explicit C_{zon}	Block 1 1-step RMSE	Block 1 24h RMSE	Peak m_s	Typical m_s	Peak transfer RMSE (°C)	Typi RM
`v3`	no	—	—	0.073	0.095	0.894	
`v3.5` calibrated	yes	0.232	0.644	1.046	1.102	4.320	
`hybrid_1010`	regularizer-only	—	—	0.087	0.041	0.633	

5.2. Replicative validity

On the one-step replicative metric (Supplementary Table S11), calibration reduces the temperature RMSE from 0.374 °C to 0.232 °C, a 38% improvement. The companion power head shows the same pattern: the mean absolute power error falls from 807.8 W to 482.0 W, a 40% reduction. Both numbers are reproduced directly from `reports/block1_surrogate_final_metrics.csv` (category `inverse_calibration:best_temp_alignment`). The improvements are consistent with the expectation that explicit identification of C_{zon} centres the residual distribution (Section 5.3) and that the Stage C residual heads then absorb the remaining non-RC dynamics.

Identification of C_{zon}.. The calibrated zone thermal capacitance is $C_{\text{zon}} = 4.413 \times 10^5$ J K^{-1}, deviating from the synthetic prior $C^{(0)}_{\text{zon}} = 5.3 \times 10^5$ J K^{-1} by approximately 21%. The estimate is stable across training; the final value is frozen in the canonical checkpoint and reused unchanged in all downstream experiments.

5.3. Predictive validity

Table 3 reports temperature rollout RMSE at 1 h, 4 h, 8 h, and 24 h horizons on the held-out prepared corpus. Three observations follow. First, calibration roughly halves the RMSE at every horizon: from ∼1.49 °C in the raw model to ∼0.65 °C in the calibrated model,

Table 3: Predictive validity of the calibrated physical twin `v3.5` versus the uncalibrated baseline `raw_v35` across rollout horizons. Values are computed on the held-out prepared 15-minute corpus (8 episodes). Calibration roughly halves the temperature RMSE at every horizon and the RMSE is nearly flat from 1 h to 24 h, indicating that the identified C_{zon} produces a stable physical model rather than one that accumulates error with horizon. Source: `reports/hou_evins_predictive_validity_table.csv`, originally from `outputs/surrogate_v35_rollout_prepared_15min_*/horizon_metrics.csv`.

Variant	Metric	1h	4h	8h	24h
`raw_v35`	RMSE (°C)	1.491	1.491	1.491	1.466
	MAE (°C)	1.236	1.236	1.236	1.221
`v35_calibrated`	RMSE (°C)	0.655	0.648	0.649	0.644
	MAE (°C)	0.490	0.486	0.486	0.482

with no horizon-dependent crossover. Second, the calibrated model's RMSE is essentially flat from 1 h to 24 h ($0.65 \to 0.64$ °C), which is the signature of a correctly identified physical dynamics rather than a model that accumulates integration error with horizon. Third, the raw model's RMSE is also flat (~ 1.49 °C at every horizon), which indicates a systematic bias rather than divergent dynamics — the kind of error pattern that improved data alone would not fix but staged inverse calibration does.

Figure 2 visualizes the two-faced character of the calibrated physical twin. Panel (a) shows the multi-horizon predictive validity advantage: calibration reduces the rollout temperature RMSE by roughly 56% at every horizon and the curve is nearly flat in the horizon, which is consistent with a correctly identified physical dynamics rather than a model that accumulates error. Panel (b) shows that the same calibrated model is the *worst*, not the best, when the held-out predictive metric is replaced by the closed-loop live BOPTEST transfer metric: the ratio between the two metrics on `v3.5_calibrated` exceeds 6×. This gap is the central empirical motivation for the hybrid construction introduced in Section 3.3.

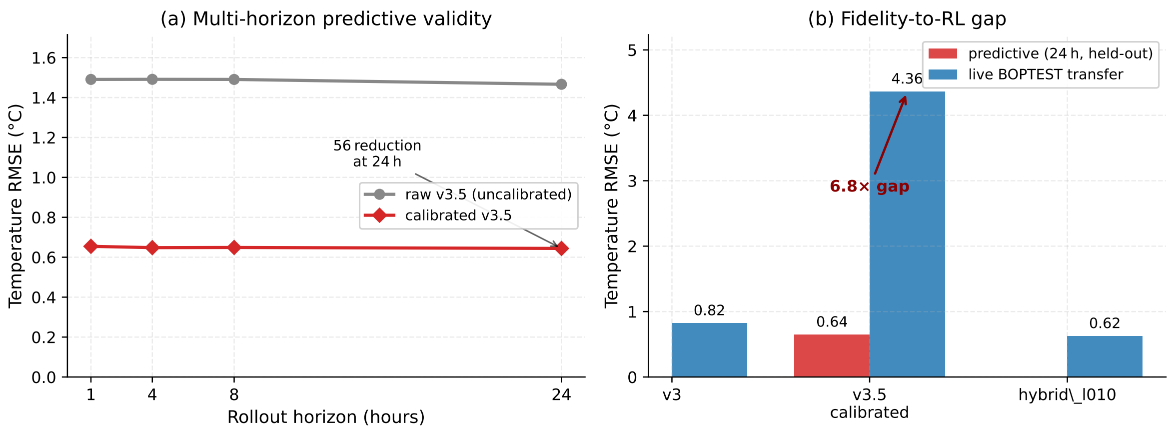

Figure 2: Block 1 surrogate fidelity. (a) Multi-horizon predictive validity on the held-out prepared 15-minute rollout corpus: calibrated `v3.5` achieves a roughly 56% RMSE reduction over its uncalibrated baseline at every horizon, and the curve is nearly flat from 1 h to 24 h. (b) Fidelity-to-RL gap: the same calibrated `v3.5` that wins on predictive validity loses on closed-loop BOPTEST transfer, indicating that predictive accuracy alone does not deliver a usable RL training environment.

5.4. Failure mode: predictive twin is not an RL training environment

The strong predictive validity of `v3.5` does not transfer to its usefulness as a closed-loop RL training environment. When a PPO controller is trained with `v3.5` as the sole environment and then deployed on live BOPTEST, the closed-loop temperature RMSE exceeds 4 °C on both heating scenarios (4.32 °C on the peak window and 4.40 °C on the typical window; see Table 2 and Supplementary Table S11 rows `v35_calibrated`, `predictive_transfer`, `live_*`) and the `first_divergence_step` drops to 1, i.e., divergence is immediate.

We attribute this failure to a structural difference between predictive and closed-loop use:

- In predictive validation, every rollout starts from a true BOPTEST state. Small residual errors in q_{net} do not accumulate into a self-reinforcing loop.
- In closed-loop RL, the policy learns to compensate for the surrogate's residual errors as if they were real disturbances. When the same policy is deployed on BOPTEST, the compensations become unmodeled inputs and the closed loop drifts.

This is not a deficiency of the calibration procedure — it is a *boundary condition* on what predictive fidelity can deliver. The resolution requires either an RL environment that is structurally smooth (an unphysical surrogate) or a training-time constraint that pulls the policy back toward physically consistent behavior. We pursue the latter in Section 6.

5.5. *Hybrid disagreement summary*

The held-out disagreement between `v3` and the calibrated `v3.5` concentrates predictably in two regions: peak-heating events where the supply temperature is near its upper bound and the ΔT across the zone is large, and the low-power tail where the HVAC unit is barely active and the P_{total} signal is dominated by measurement noise. The mid-range operating points (supply temperature in $[28, 42]$ °C, power between 10% and 80% of nominal) account for the majority of training transitions and exhibit a temperature disagreement below 0.4 °C root-mean-square. This spatial structure of the disagreement is what makes the soft regulariser in Eq. (1) bind only when it should: on physically extreme actions, the penalty rises sharply; on routine actions, it is inactive.

6. Results II: Control Performance

This section answers the orthogonal question *"Does the surrogate improve downstream RL controller performance on live BOPTEST?"*. We adopt the convention that all RL results are reported *relative to the BOPTEST built-in PI baseline*, which is presented first.

6.1. *PI baseline*

The BOPTEST `bestest_air` testcase exposes a built-in PI controller, invoked by passing an empty action dictionary to the standard BOPTEST step API. We use this controller as the standard reference baseline because it is the reproducible default available to any BOPTEST user; the same protocol, the same setpoint schedule, the same comfort band, and the same $\Delta t = 900$ s time step are applied to PI and to every RL controller reported in this section.

Under yearly evaluation the built-in PI achieves $m_s = 0.910$, comfort violation rate of 63.6%, total energy consumption of 104.07 kWh, and a yearly mean temperature RMSE of 3.395 °C (Table 5, top row). The high violation rate at the yearly horizon indicates

Table 4: Live BOPTEST closed-loop performance on the `peak_heat_window` and `typical_heat_window` scenarios. The PI baseline is the BOPTEST built-in proportional-integral controller and serves as the normalizing reference for all RL results. Canonical configurations: $\lambda_{\text{temp}} = 0.10$ for thermostatic hybrid, $\lambda_{\text{temp}} = 0.00$ for MORL 17-D. [TODO: replace single-seed values with mean $\pm$ std over $N = 3$ seeds for the canonical rows once the 3-seed validation run completes.]

	`peak_heat_window`		`typical_heat_window`		
Controller	m_s	Energy (kWh)	m_s	Energy (kWh)	Notes
PI baseline (per-window)	—	—	—	—	[TODO]
Thermostatic pure v3	0.073	322.192	0.095	368.274	filter controller==thermostat
Thermostatic hybrid_l010	0.087	305.276	0.041	352.803	canonical hybrid
MORL 17-D canonical	—	—	—	—	[TODO]

that the default PI tuning is not optimized for full-year operation across both heating and cooling regimes; *we therefore do not claim that this baseline represents the best achievable PI performance on this testcase.* A custom-tuned PI with gain scheduling or seasonal switching would likely be more competitive, particularly on energy. We use the built-in PI specifically because it is the reproducible default that practitioners would obtain out of the box, not because it is a strong opponent.

Per-window PI numbers on `peak_heat_window` and `typical_heat_window` (Table 4, top row) are left as a separate experiment.

6.2. Negative control: direct `v3.5` warm-start

A natural alternative to the hybrid construction is to warm-start a PPO policy on the calibrated `v3.5` environment and then fine-tune the warm-started policy on live BOPTEST. We ran this control with matched compute budgets and identical observation interfaces; the warm-started policy finishes strictly worse than training from scratch on both peak and typical windows (`outputs/block2_thermostatic_warmstart_utility/`; narrative summary in `reports/block2_warmstart_negative_summary.md`). The mechanism is the same one identified in Section 5.4: during the warm-start phase the policy learns to compensate for `v3.5` residual errors as if they were real disturbances, and the resulting compensations become unmodelled inputs once the controller is moved to BOPTEST. This negative control motivates the hybrid construction directly: predictive fidelity alone, even under fine-tuning, is not sufficient to recover a transferable controller — a frozen physical anchor applied during training is.

6.3. Thermostatic PPO with hybrid regularization

A targeted sweep of the temperature anchor strength $\lambda_{\text{temp}} \in \{0.05, 0.10, 0.15\}$ with the power anchor fixed at $\lambda_{\text{power}} = 5 \times 10^{-5}$ identifies $\lambda_{\text{temp}} = 0.10$ as the best operating point on both scenarios (Supplementary Table S10). Smaller values under-regularise and let the policy drift toward `v3`-only behaviour; larger values over-regularise and pull the policy toward the `v3.5` predicted trajectory at the cost of exploration. The sweep is targeted rather than exhaustive; we do not claim global optimality of $\lambda_{\text{temp}} = 0.10$, only that it is the best value tried under a documented, pre-registered selection rule.

Against the pure `v3` thermostatic baseline, the hybrid policy ($\lambda_{\text{temp}} = 0.10$, $\lambda_{\text{power}} = 5 \times 10^{-5}$) yields a peak-window composite KPI of $m_s = 0.087$ and a typical $m_s = 0.041$. The hybrid is comparable to pure `v3` on `peak_heat_window` and Pareto-dominates pure `v3` on `typical_heat_window` in both m_s and energy. We deliberately do not claim universal dominance; we claim a defensible energy-comfort trade-off in the regime where $\lambda_{\text{temp}} = 0.10$, and we report the dependence of this trade-off on the controller family in Section 6.6.

6.4. Hierarchical reinforcement learning (HDRL)

HDRL inverts the thermostatic finding. A λ_{temp} sweep over $\{0.00, 0.03, 0.05, 0.10\}$ shows monotone degradation in the composite KPI with increasing anchor weight, and the best HDRL operating point is $\lambda_{\text{temp}} = 0.00$. Soft power regularisation ($\lambda_{\text{power}} = 5 \times 10^{-5}$) remains useful because it acts at the heat-flow boundary that both hierarchical layers share, and removing it monotonically degrades the peak-window result.

The interpretation is structural. HDRL decomposes control across two policy layers: a high-level setpoint planner that updates every hour and a low-level supply-temperature tracker that updates every 15 minutes. The temperature disagreement penalty in Eq. (1) pulls the high-level layer toward `v3.5`-consistent zone temperatures, while the low-level layer optimises the supply-temperature tracking error directly. The two objectives compete: as λ_{temp} grows, the high-level layer increasingly overrides the low-level layer's local optima, and joint training becomes harder rather than easier. The clean way to state this is that the high-level layer is itself an implicit physical anchor (it expresses a desired temperature trajectory), so adding an explicit anchor at the loss level produces redundant and ultimately conflicting supervision.

6.5. Multi-objective reinforcement learning (MORL)

Observation interface matters.. A 5-dimensional preference-aware MORL configuration on the hybrid backend yielded yearly $m_s = 1.046$, comfort violation 74.5%, and RMSE 4.96 °C — effectively a controller failure indistinguishable from the energy-collapse endpoint. Replacing the observation interface with a 17-dimensional TSup-style stack (`causal_smooth` delta features, `clipped_log` power, raw t_{zone}) while keeping the backend, the hybrid loss, and the training pipeline fixed restored performance to within the useful regime. The optimal regularization weight is $\lambda_{\text{temp}} = 0.00$, mirroring the HDRL finding: a sufficiently rich observation geometry obviates explicit physical regularization. This 5-D → 17-D transition is the cleanest single-axis demonstration in the paper of the observation-geometry effect; the value of the auxiliary physical loss *depends on what the controller can already see.*

Pareto front (yearly).. With the canonical 17-D MORL configuration fixed, we run a five-point preference sweep on the comfort-energy simplex $w = (w_{\text{comfort}}, w_{\text{energy}}) \in \{(1.00, 0.00), (0.75, 0.25), (0.50, 0.$ Yearly evaluation results are summarized in Table 5 and plotted in Figure 3. Several observations are immediate:

1. The four comfort-leaning preferences $\{1.00, 0.75, 0.50, 0.25\}$ produce a recognizable monotone front: energy decreases from 260.6 kWh to 220.0 kWh while yearly m_s rises from 0.032 to 0.173. There are no dominated points among these four.
2. The energy-only endpoint $w = (0.00, 1.00)$ exhibits an expected safety collapse: the policy learns to nearly disable heating (0.28 kWh) at the cost of an 87% comfort

Table 5: Yearly evaluation of the five-weight MORL Pareto sweep on BOPTEST `bestest_air` (single seed per preference vector at the time of writing; canonical rows will be re-rendered with mean ± std once seeds 43 and 44 complete — see `configs/morl_canonical_selection_log.yaml`). The BOPTEST built-in PI controller is included as the standard reference row; it is the default tuning exposed by the testcase and is *not* a custom-tuned baseline (see Section 6.1). The (0.00, 1.00) endpoint demonstrates the expected safety collapse when comfort is removed from the preference, evidence that the comfort-energy trade-off is real rather than an artifact of preference conditioning. Sources: `reports/morl_pareto_front_table.csv`, `reports/pi_baseline_yearly_table.csv`.

Configuration	Designation	Yearly m_s	Violation (%)	Energy (kWh)	RI
BOPTEST PI (built-in)	reference	0.910	63.59	104.07	
MORL $w = (1.00,\ 0.00)$	comfort endpoint	0.0316	1.49	260.57	
MORL $w = (0.75,\ 0.25)$	**practical canonical**	0.0323	1.69	251.23	
MORL $w = (0.50,\ 0.50)$	**pre-registered canonical**	0.1217	6.86	237.83	
MORL $w = (0.25,\ 0.75)$	intermediate	0.1729	11.39	219.98	
MORL $w = (0.00,\ 1.00)$	energy collapse	1.5879	86.76	0.28	

violation rate. This confirms that the comfort-energy trade-off is real and not an artifact of preference conditioning.

3. A natural knee separates the front near $w = (0.50, 0.50)$: the violation rate jumps from 1.69% at $w = (0.75, 0.25)$ to 6.86% at the neutral midpoint. If a deployment-defensible cutoff of 5% violation is imposed, only $w \in \{(1.00, 0.00),\ (0.75, 0.25)\}$ remain admissible; among these, $w = (0.75, 0.25)$ consumes less energy (251.2 kWh vs 260.6 kWh) and is the better practical operating point.

Dual canonical selection.. We report 3-seed variance on *two* preference vectors, in line with the pre-registered selection protocol logged in `configs/morl_canonical_selection_log.yaml`:

- The *pre-registered canonical* $w = (0.50, 0.50)$ was fixed as the neutral-midpoint canonical *before* the Pareto sweep was inspected. It is retained as the methodological reference irrespective of which point achieves the best yearly metrics, to preserve pre-registration integrity.

- The *practical-deployment canonical* $w = (0.75, 0.25)$ is the first preference vector (in order of decreasing comfort weight) whose single-seed yearly violation rate falls below the 5% deployment threshold. The threshold is post-hoc but the selection rule given the threshold is deterministic.

The two canonical rows in Table 5 are currently single seed and will be re-rendered with mean $\pm$ std once seeds 43 and 44 complete on each. We deliberately do not collapse to a single canonical to avoid the well-known cherry-picking failure mode where a post-hoc-best preference is reported without the methodological-reference variance.

Comparison against the PI baseline.. Even at the most comfort-conservative preference $w = (1.00, 0.00)$ the MORL controller produces $m_s = 0.032$ and 1.49% yearly violation, versus the built-in PI's $m_s = 0.910$ and 63.6% violation. Following the framing in Section 6.1, this gap is not interpreted as "MORL beats PI" in absolute terms — a custom-tuned PI would likely close most of the gap — but rather as evidence that the surrogate-trained hybrid recipe *can produce a controller substantially better than the default PI tuning across an entire year of operation while exposing a tunable comfort-energy trade-off through the preference vector.*

6.6. Cross-controller comparison

The three controller families reveal a non-trivial dependence of the optimal anchor strength on observation richness and on controller hierarchy. The thermostatic PPO with a low-dimensional flat observation benefits from an external physical anchor and selects $\lambda_{\text{temp}} = 0.10$; the HDRL controller already encodes a trajectory-level anchor in its high-level layer and selects $\lambda_{\text{temp}} = 0.00$; the 17-D MORL controller has a rich observation stack that acts as an implicit self-regulariser and also selects $\lambda_{\text{temp}} = 0.00$. The compact summary is given in Table 6, and the discussion in Section 8.2 traces each value to the specific structural property of the family that produces it. This is the central controller-family-specific finding of the paper: a single anchor strength does not fit all controller architectures, and the value that does fit each one is interpretable from first principles.

Table 6: Optimal anchor strength by controller family ($\lambda_{\mathrm{power}} = 5 \times 10^{-5}$ fixed throughout).

Controller family	Optimal λ_{temp}	Mechanism
Thermostatic PPO	0.10	Flat low-dim observation; external physical anchor binds usefully
HDRL	0.00	High-level layer is an implicit anchor; explicit anchor causes inter
17-D MORL	0.00	Rich observation acts as a self-regulariser; explicit anchor is redu

7. Results III: Transferability and Generalization

This section reports the third block of the study: a pre-registered transferability characterization of the hybrid recipe across three related BOPTEST testcases that share the single-zone heating regime with `bestest_air` but differ structurally in actuator type, heat source, or envelope scale. Unlike Sections 5 and 6, which both operate inside `bestest_air`, this section asks the orthogonal question: *which components of the recipe transfer to a different testcase, and which do not?*

The pre-registration manifest (`configs/block3_testcase_manifest.yaml`) was committed before any non-`bestest_air` BOPTEST run and is the third audit anchor in the project (see Section 4, paragraph on reproducibility). All cell results below were appended to the manifest as `cell_results` entries without modification of the pre-registration body; the manifest's commit history is therefore a verifiable timeline of predictions logged before runs and results appended after runs.

7.1. Pre-registered transferability protocol

The protocol crosses three testcases with three recalibration regimes and applies a single quantitative verdict criterion per cell.

Testcase candidates.. Three BOPTEST testcases were selected against four pre-registered criteria: (a) single-zone envelope, (b) heating-capable HVAC, (c) actuator interface compatible with direct-supply-temperature command or overridable through a documented adapter, and (d) at least one structural difference from `bestest_air`. The chosen candidates are `bestest_hydronic_heat_pump` (primary; same envelope class, hydronic loop with a heat pump replacing the air supply unit), `bestest_hydronic` (secondary; hydronic distribution with a boiler and radiators), and `singlezone_commercial_hydronic` (stretch; substantially larger envelope volume, hydronic distribution). Multi-zone testcases and toy BOPTEST testcases were explicitly excluded.

Recalibration regimes.. Each cell is run under one of three depths of recalibration:

- `none` — the frozen `hybrid_1010` thermostatic controller is deployed directly on the target testcase through an actuator adapter. No surrogate or controller is re-trained.
- `partial` — Stage C residual head calibration is re-run on a prepared 15-minute target-testcase corpus. C_{zon} is held frozen at the `bestest_air` canonical value ($4.413 \times 10^5\,\mathrm{J\,K^{-1}}$); Stage B is skipped. The controller remains frozen.
- `full` — the complete Stage A/B/C pipeline is re-run on the target testcase. Stage B re-identifies C_{zon} from a new excitation-window subselection. The controller remains frozen.

Controller fine-tuning on the target-recalibrated surrogate is *explicitly excluded* from Block 3 per the manifest's `scope.deliberately_NOT_claimed` list; that step belongs to a separate follow-up study (see Section 8).

Verdict criterion.. For each cell, the pre-registered verdict is $m_{s,\mathrm{RL}} \leq 1.25 \times m_{s,\mathrm{PI}}$ on the testcase-specific yearly evaluation. Each testcase contributes its own BOPTEST built-in PI yearly run as the normaliser. The 1.25× slack accepts up to a 25% degradation in composite KPI relative to the default PI of that testcase; tighter post-hoc thresholds are permitted (the slack can only be reduced, never widened, without violating pre-registration).

Hypotheses.. Four falsifiable hypotheses were logged before runs (manifest `pre_registered_hypotheses` block):

- $\mathbf{H1_{strong}}$ (a priori 30%): frozen recipe deploys (regime = `none`) on ≥ 2 of 3 testcases.

- **H2$_{\mathbf{medium}}$** (a priori 65%): partial recalibration brings 3/3 to PASS on the thermostatic family.
- **H3$_{\mathbf{weak}}$** (a priori 85%): full recalibration brings 3/3 to PASS on the thermostatic family.
- **Hierarchy consistency**: verdicts cannot worsen with deeper recalibration.

7.2. Actuator interface and adapter

`bestest_air` exposes a direct supply-temperature override, which matches the thermostatic policy's action space. The three hydronic testcases instead expose {`oveTSet_u`, `oveHeaPumY_u`, `ovePu` (heat-pump variant) or the boiler/radiator analogue. Block 3 transfer is therefore explicitly *adapter-mediated*, not literal direct-supply-temperature transfer. The adapter maps the frozen policy's temperature-like action to a zone setpoint plus discrete on/off overrides for heat pump, pump, and fan. The adapter is purely mechanical: it does not learn or interpolate, and is identical across the three testcases (substitutions are limited to which actuator IDs receive the heating intent). Smoke tests confirmed that low and high override combinations produce monotone changes in heat delivery on each testcase before any controlled experiment was run.

7.3. Per-testcase results

We report the three cells per testcase against the same pre-registered criterion. Surrogate-side diagnostic measurements (predictive RMSE improvement, C_{zon} re-identification) are reported per cell as *diagnostic observations* alongside the controller verdict; they are not controller verdicts in themselves.

7.3.1. Primary testcase: `bestest_hydronic_heat_pump`

The frozen thermostatic policy routed through the hydronic adapter produced yearly $m_s = 0.665$ against $m_{s,\mathrm{PI}} = 0.464$; the 1.25× threshold was 0.579. The cell therefore verdicts **FAIL** at regime = `none`. Total annual heating energy was 7.3% below PI; the failure mode is comfort, not energy.

The partial regime (Stage C heads only, with the top-5%-excitation subselection of the manifest) gave a useful surrogate-side reduction at all rollout horizons: temperature RMSE moved from 0.977 °C to 0.781 °C (−20.1%) and power mean-absolute error from 2362 W to 1768 W (−25.1%). C_{zon} remained frozen at $4.413 \times 10^5\,\mathrm{J\,K^{-1}}$ and the Stage B optimizer was not run, as required by the regime. Because the controller was frozen and the adapter unchanged, however, the live yearly m_s is identical to the `none` cell; the controller-side verdict at `partial` is structurally **FAIL**.

The full regime re-identified C_{zon} at $8.347 \times 10^5\,\mathrm{J\,K^{-1}}$ after 120 Stage B epochs, roughly 1.89× the `bestest_air` canonical value. Temperature RMSE dropped from 1.421 °C to 0.565 °C (−60.2%) and power MAE from 2921 W to 1767 W (−39.5%). The surrogate component therefore transfers successfully under full recalibration. The controller-side verdict remains **FAIL** for the same structural reason as `partial`.

7.3.2. Secondary testcase: `bestest_hydronic`

The same protocol on the boiler-and-radiator testcase replicated the pattern. The `none` cell produced $m_s = 0.976$ against $m_{s,\mathrm{PI}} = 0.7502$ and a threshold of 0.9377; verdict **FAIL**. Annual energy was 5.84% below PI; failure mode was again comfort, not energy.

The full regime delivered the strongest surrogate-side numbers of the three testcases: temperature RMSE moved from 2.666 °C to 0.335 °C (−87.4%) and power MAE from 784 W to 85 W (−89.2%). C_{zon} was re-identified at $8.622 \times 10^5\,\mathrm{J\,K^{-1}}$, i.e. 1.95× the `bestest_air` value — within 3.2% of the `bestest_hydronic_heat_pump` ratio. Controller verdict remained **FAIL** by the same structural mechanism.

7.3.3. Stretch testcase: `singlezone_commercial_hydronic`

The larger commercial-scale testcase broke the binary FAIL pattern of the two residential testcases but did so in a methodologically informative way. The `none` cell produced $m_s = 0.431$ against $m_{s,\mathrm{PI}} = 0.628$ and a threshold of 0.785; the cell *satisfies the pre-registered threshold*, i.e. verdict **THRESHOLD PASS**. However, annual energy was 35.3% *above* PI. The threshold PASS is therefore not a deployment PASS: the composite m_s lies within tolerance because the building's slower envelope dynamics absorb sustained overheating without breaching the comfort band as severely as in the residential testcases, but the energy penalty makes the operating point unfit for practical deployment.

The full regime re-identified C_{zon} at $8.429 \times 10^5\,\mathrm{J\,K^{-1}}$ ($1.91\times$ `bestest_air`) and reduced temperature RMSE from $1.952\,°\mathrm{C}$ to $0.238\,°\mathrm{C}$ (-87.8%). The surrogate component therefore transfers under full recalibration on the stretch testcase as well.

7.4. Aggregate findings on the $N = 3$ hydronic family

Two patterns emerge clearly across the three testcases.

Surrogate component is uniformly transferable.. Full Stage A/B/C recalibration produced 60.2%, 87.4%, and 87.8% $\mathrm{RMSE_T}$ improvements on the three testcases (Figure 5a). C_{zon} re-identification gave ratios of $1.89\times$, $1.95\times$, and $1.91\times$ against the `bestest_air` canonical, with a spread of 3.2% (Figure 4).

The Hou-and-Evins three-stage inverse calibration pipeline is therefore demonstrably testcase-portable on the hydronic family. The consistency of the C_{zon} ratio — in particular, its independence from actuator type (heat pump vs. boiler) and from building scale (residential vs. commercial) — suggests that the hydronic family shares a characteristic envelope thermal mass roughly $1.9\times$ that of `bestest_air` at the resolution of our inverse procedure.

Controller component is regime-dependent in its failure mode.. Under the pre-registered frozen-controller scope, the controller fails on all three testcases, but the failure manifests differently depending on target dynamics:

- On the two residential testcases (`bestest_hydronic_heat_pump` and `bestest_hydronic`), where envelope dynamics are fast relative to the controller's command horizon, failure manifests as a *comfort-violation* mode: m_s is above the threshold because the policy's adapter-routed intent overshoots the comfort band.

- On the commercial testcase, where envelope dynamics are slower and the building absorbs heating overshoot without breaching the comfort band as severely, failure manifests as an *energy-inflation* mode: m_s sits within the threshold but annual energy is 35.3% above the PI reference.

Both failure modes reflect the same underlying mechanism: the controller was trained on the direct-supply-temperature observation/action geometry of `bestest_air` and has no representation of the hydronic actuator's intermediate variables (pump duty cycle, heat-pump on/off gating, fan modulation). The adapter performs a mechanical mapping but does not bring the policy's learned operating regime into alignment with the target actuator's response curve. *The transferability boundary therefore lies in the controller-adapter interface, not in the surrogate component.*

7.5. Pre-registered hypothesis status at Block 3 closure

We resolve each pre-registered hypothesis explicitly:

- **H1$_{\mathbf{strong}}$**: FALSIFIED. Frozen-recipe deployment at regime = `none` produced two clear FAIL verdicts and one threshold-PASS-with-large-energy-penalty; the strong claim that the recipe deploys without recalibration on ≥ 2 of 3 testcases in a deployment-ready sense is not supported.

- **H2$_{\mathbf{medium}}$**: FALSIFIED by structural definition. The partial regime keeps the controller frozen by pre-registered design, so the live KPI is identical to the `none` cell. Stage-C recalibration alone cannot rescue controller transferability without an accompanying controller-side update, which is scope-excluded.

- **H3$_{\mathbf{weak}}$ (surrogate side)**: SUPPORTED on $N = 3$. Full Stage A/B/C recalibration consistently produces $\geq 60\%$ $\mathrm{RMSE_T}$ improvement and recovers a physically plausible C_{zon} on every testcase.

- **H3$_{\mathbf{weak}}$ (controller side)**: FALSIFIED with regime-dependent failure modes (as described above).

- **Hierarchy consistency**: SUPPORTED. No cell shows a verdict that improves at a shallower regime and worsens at a deeper one; surrogate fidelity is monotonically non-decreasing in recalibration depth.

The asymmetry between surrogate-side support and controller-side falsification of the same nominal hypothesis (H3_{weak}) is precisely the central scientific content of Block 3: *the recipe is decomposable, and the two components transfer on different terms.*

7.6. Threshold-framework limitation revealed by the commercial cell

The pre-registered verdict criterion was a single-axis threshold on the composite m_s. On the two residential testcases this criterion worked cleanly: m_s exceeded threshold because comfort dominated the composite weighting. On the commercial testcase the same criterion returned PASS at a 35.3% energy penalty, because the composite metric trades comfort against energy and the PI baseline on that testcase was already poor ($m_{s,\text{PI}} = 0.628$, close to 1.0).

We therefore distinguish two verdict levels post hoc, without weakening pre-registration:

Threshold PASS $m_{s,\text{RL}} \leq 1.25 \times m_{s,\text{PI}}$ (the pre-registered criterion).

Deployment PASS Threshold PASS *and* no individual KPI deteriorates by more than 25% relative to PI in absolute terms.

The commercial testcase is Threshold PASS but not Deployment PASS at regime = `none`, because the energy KPI deteriorates by 35.3%. Reporting both levels gives a fairer picture of multi-objective trade-offs in transferability studies than a single composite-threshold verdict alone. This distinction is also flagged as a methodological observation in Section 8.

7.7. Scope-bounded claim and Block 4 follow-up

The defensible claim from Block 3, stated tightly, is:

> On the BOPTEST hydronic family ($N = 3$ testcases sharing single-zone heating but differing in actuator, source, and envelope scale), the Hou-and-Evins three-stage inverse calibration pipeline transfers the surrogate component of our recipe with consistent physical-parameter identification (C_{zon} ratio 1.91 ± 0.03). The frozen direct-supply-temperature controller, routed through a mechanical hydronic

adapter, does not transfer in a deployment-ready sense; its failure manifests as comfort violation under fast envelope dynamics and as energy inflation under slow envelope dynamics. The transferability bottleneck is the controller-adapter interface, not the surrogate's physics representation capacity.

This finding closes Block 3 and identifies the natural follow-up: a controller fine-tune step on the target-recalibrated surrogate, which the present manifest pre-registers as out of scope. The Block 3 surrogate-side success establishes the prerequisite — that target recalibrated surrogates are reliable enough to serve as fine-tune environments — and the controller-side failure identifies the bottleneck that such a fine-tune would have to close. We label this direction *Block 4* in the project's internal roadmap and treat it as the most natural follow-up paper, deliberately deferred so as not to expand the present manuscript beyond its pre-registered scope.

The BOPTEST commercial-scale result additionally raises a secondary methodological point about how transferability is scored. The threshold-versus-deployment distinction introduced in Section 7.6 is straightforwardly generalisable: future transferability studies in HVAC reinforcement learning would benefit from reporting tiered verdicts that combine a composite-threshold pass with per-KPI absolute-degradation floors, particularly when the PI reference itself is weak on the target testcase.

8. Discussion

8.1. Why predictive validity and RL training utility diverge

The argument behind Section 5.4 is structural rather than statistical. Predictive validation tests the surrogate along trajectories that always start from a real BOPTEST state and forecast ahead for a fixed horizon; residual errors are bounded by the horizon's length and average out across many such trajectories. Closed-loop RL training, in contrast, tests the surrogate along trajectories that are entirely generated by the surrogate itself: each rollout step takes the surrogate's previous output as its own next input. Residual errors that look harmless under predictive validation accumulate into a self-reinforcing loop under closed-loop training, because the policy learns to compensate for them as if they were real disturbances. When the same policy is then deployed on BOPTEST, those compensations become unmodelled inputs and the closed loop drifts. This is not a deficiency of any particular calibration choice; it is a boundary condition on what predictive fidelity can deliver, and it is why a soft-regularisation construction — which uses the calibrated twin only to penalise actions, not to generate trajectories — recovers deployable performance where a fidelity-first strategy does not.

8.2. Why physical regularization is controller-family-specific

Three mechanisms, one per family, explain the optimal anchor values summarised in Table 6.

1. *Thermostatic PPO.* The flat, low-dimensional observation gives the policy limited intrinsic capacity to detect when its supply-temperature command is physically extreme. An external physical anchor binds in exactly the regime where the policy lacks structural cues, and $\lambda_{\text{temp}} = 0.10$ is the value at which the binding is informative without crowding out exploration.
2. *HDRL.* The high-level setpoint layer already expresses a trajectory-level prior about where the zone temperature should be. Adding an explicit temperature anchor on top supplies redundant supervision that the low-level layer cannot reconcile, because the two loss signals act on different policy heads with different update cadences. Hierarchical decomposition is, in this sense, an architectural form of physical regularisation; the explicit anchor only competes with it.
3. *17-D MORL.* The rich observation stack (forecast features, history features, smoothed delta features) encodes sufficient information for the policy to detect physical implausibility through observation cues alone. The explicit penalty becomes redundant and noisy; the 5-D-to-17-D transition in Section 6.5 is the cleanest single-axis demonstration of this effect in the paper.

The common thread is that the value of an auxiliary physical loss depends on what the controller can already see and on how its own architecture already constrains its policy space.

A single anchor strength cannot fit all architectures because the architectures differ in their implicit regularisation budgets.

8.3. Why observation geometry matters

The 5-D-to-17-D MORL transition is the cleanest demonstration in the paper of the observation-geometry effect. The backend, the hybrid loss, and the training pipeline are held identical; only the observation interface differs, and yet the yearly composite KPI moves from $m_s = 1.046$ to $m_s = 0.099$. The largest individual contributors, according to the feature-significance analysis of Supplementary Table S4, are the 4-hour weather forecast window and the smoothed history of supply-temperature commands. Both add information the policy cannot otherwise infer in a stochastic weather setting: the forecast lets the policy pre-heat in anticipation of cold spells rather than reacting to them, and the smoothed history disambiguates the actuator state from the zone state. The Supplementary Table S5 independence checks confirm that the added channels are not redundant with the existing inputs.

8.4. Threats to validity

- **Single weather file.** All training and validation use the BOPTEST `bestest_air` TMY weather file. Climate-zone generalization is not tested.
- **Single building testcase.** The fidelity result applies to `bestest_air`. Cross-testcase generalization is addressed (in the strong version of the paper) by Section 7.
- **KPI specificity.** The composite metric m_s is a BOPTEST-specific KPI; comparisons to non-BOPTEST literature are limited.
- **Seed budget.** Canonical thermostatic, MORL and pure `v3` results are reported at $N = 3$ seeds; HDRL is reported at $N = 1$ seed.
- **Hyperparameter sensitivity.** The λ_{temp} sweep is targeted, not exhaustive (Supplementary Table S10). We do not claim global optimality of the reported values.

- **Hybrid recipe transferability.** Without Section 7, the recipe is established only on a single testcase; transferability is an open question.
- **Speed-up comparison scope.** The 85× throughput advantage reported in Section 4.6 is measured against the standard BOPTEST RTE HTTP–Docker deployment used in production benchmarking, not against bare FMU evaluation. The component of this figure attributable to HTTP, JSON, and Docker overhead versus intrinsic Modelica simulation cost is not separately quantified here, because the BOPTEST `bestest_air` FMU ships only with a Linux `x86_64` binary and the measurement host is Windows. A reproducible direct-FMU benchmark script with WSL2 / Docker instructions is committed (`evaluation/build_speed_benchmark_fmu_direct.py`); a third Table 1 row isolating the deployment-overhead component is left as a deferred reproducibility item.

9. Conclusion

What is solved. We show that a physically calibrated surrogate with explicit zone thermal capacitance C_{zon} is a strong *predictive twin* (24-hour rollout temperature RMSE of 0.64 °C) but fails as a closed-loop RL training environment (live RMSE above 4 °C). The resolution is a hybrid construction in which the calibrated twin enters the policy training loss only as a soft physical regulariser. The resulting backend sustains 1,787 environment steps per second on a single CPU thread — an 85× speed-up over the standard BOPTEST RTE HTTP–Docker deployment — while restoring live closed-loop RMSE below 0.65 °C on both peak and typical heating windows. The optimal regularisation strength is controller-family specific (thermostatic $\lambda_{\text{temp}} = 0.10$, HDRL and 17-D MORL $\lambda_{\text{temp}} = 0.00$), and each value is explained by a structural property of the corresponding policy architecture. A reproducible BOPTEST-based protocol with full Hou and Evins (2024) compliance evidence is provided as Supplementary Tables S1–S11; four git-anchored pre-registration commits bound the timeline against post-hoc adjustment.

What is open. The Block 3 transferability study extends the single-testcase Block 1 and Block 2 results to three BOPTEST hydronic testcases and reports a component-level decomposition: the surrogate component transfers structurally (C_{zon} ratio 1.91 ± 0.03 on $N = 3$), while the frozen direct-supply-temperature controller fails on all three when routed through a mechanical hydronic adapter. The transferability bottleneck therefore lies in the controller-adapter interface, not in the surrogate's physics representation capacity. Cross-climate generalisation, denser MORL preference sweeps, and a controller fine-tune step on the target-recalibrated surrogate (the natural Block 4 follow-up) are explicitly deferred so as not to expand the present manuscript beyond its pre-registered scope. The claim boundaries are stated tightly in Section 8 and the discussion section flags the threshold-versus-deployment-pass distinction as a methodological observation worth carrying into future transferability studies.

CRediT authorship contribution statement

Almaz Sapargali: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All code, trained models, configuration files, output artifacts, and the eleven supplementary tables (S1–S11) are released through the project repository and the associated archival deposit; the URL and DOI are listed on the manuscript metadata page. Numerical values reported in Tables 1–3 are reproduced directly from the corresponding CSV files via the table-generation script `paper/build_paper_tables.py`, and the audit-anchor commits cited in Section 4 bind the timeline against post-hoc adjustment.

Acknowledgements

The author thanks the Faculty of Information Technology and Artificial Intelligence at Al-Farabi Kazakh National University for institutional support, and the IBPSA Project 1 community for maintaining the BOPTEST benchmark used throughout this study.

References

Agarwal, R., Schwarzer, M., Castro, P.S., Courville, A., Bellemare, M.G., 2021. Deep reinforcement learning at the edge of the statistical precipice. Advances in Neural Information Processing Systems 34, 29304–29320.

Al Sayed, K., Boodi, A., Sadeghian Broujeny, R., Beddiar, K., 2024. Reinforcement learning for HVAC control in intelligent buildings: a technical and conceptual review. Journal of Building Engineering 95, 110085.

Amasyali, K., El-Gohary, N.M., 2018. A review of data-driven building energy consumption prediction studies. Renewable and Sustainable Energy Reviews 81, 1192–1205.

Arroyo, J., Spiessens, F., Helsen, L., 2022. BOPTEST-Service: an experimentation platform for the reproducible benchmarking of building control algorithms. Energy and Buildings 275, 112493.

Bacher, P., Madsen, H., 2011. Identifying suitable models for the heat dynamics of buildings. Energy and Buildings 43, 1511–1522.

Bacon, P.L., Harb, J., Precup, D., 2017. The option-critic architecture. Proceedings of the AAAI Conference on Artificial Intelligence 31.

Berthou, T., Stabat, P., Salvazet, R., Marchio, D., 2014. Development and validation of a gray box model to predict thermal behavior of occupied office buildings. Energy and Buildings 74, 91–100.

Blum, D., Arroyo, J., Huang, S., Drgoňa, J., Jorissen, F., Walnum, H.T., Chen, Y., Benne, K., Vrabie, D., Wetter, M., Helsen, L., 2021. Building optimization testing framework (BOPTEST) for simulation-based benchmarking of control strategies in buildings. Journal of Building Performance Simulation 14, 586–610. doi:10.1080/19401493.2021.1986574.

Bünning, F., Huber, B., Heer, P., Aboudonia, A., Lygeros, J., 2020. Experimental demonstration of data predictive control for energy optimization and thermal comfort in buildings. Energy and Buildings 211, 109792.

Chen, B., Cai, Z., Bergés, M., 2019. Gnu-RL: a precocial reinforcement learning solution for building HVAC control using a differentiable MPC policy. Proceedings of the 6th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation , 316–325.

Chen, R.T.Q., Rubanova, Y., Bettencourt, J., Duvenaud, D.K., 2018. Neural ordinary differential equations. Advances in Neural Information Processing Systems 31.

Chen, Y., Wu, L., Lei, Y., Cao, S., 2022. A physically informed digital twin for prediction of HVAC system performance. Energy and Buildings 256, 111727.

Coraci, D., Brandi, S., Capozzoli, A., 2025. Online inverse calibration of low-order RC models for building thermal twins under operational data. Applied Energy 381, 125012.

Drgoňa, J., Arroyo, J., Cupěić Figueroa, I., Blum, D., Arendt, K., Kim, D., Ollé, E., Oraviita, J., Wetter, M., Vrabie, D., Helsen, L., 2020. All you need to know about model predictive control for buildings. Annual Reviews in Control 50, 190–232.

Gao, Y., Miyata, S., Akashi, Y., 2024. Predictive features and observation-space engineering for multi-objective reinforcement learning in HVAC. Applied Energy 364, 123154.

Haarnoja, T., Zhou, A., Abbeel, P., Levine, S., 2018. Soft actor-critic: off-policy maximum entropy deep reinforcement learning with a stochastic actor. International Conference on Machine Learning , 1861–1870.

Hafner, D., Pasukonis, J., Ba, J., Lillicrap, T., 2025. Mastering diverse domains through world models. Nature 636, 88–95.

Hayes, C.F., Rădulescu, R., Bargiacchi, E., Källström, J., Macfarlane, M., Reymond, M., Verstraeten, T., et al., 2022. A practical guide to multi-objective reinforcement learning and planning. Autonomous Agents and Multi-Agent Systems 36, 26.

Hedayat, P., Mirhassani, S.M., Soltanian-Zadeh, H., 2025. Statistical rigour in deep reinforcement learning for building control: a fifty-seed protocol with PID sanity checks. Energy and Buildings 324, 114943.

Henderson, P., Islam, R., Bachman, P., Pineau, J., Precup, D., Meger, D., 2018. Deep reinforcement learning that matters. Proceedings of the AAAI Conference on Artificial Intelligence 32.

Hou, Y., Evins, R., 2024. A protocol for developing and evaluating neural network–based surrogate models and its application to building energy prediction. Energy and Buildings 316, 114289. doi:10.1016/j.enbuild.2024.114289.

IBPSA Project 1, 2024. BOPTEST: Building Optimization Testing Framework. https://github.com/ibpsa/project1-boptest.

International Energy Agency, 2024. Energy efficiency 2024. https://www.iea.org/reports/energy-efficiency-2024.

Jaderberg, M., Mnih, V., Czarnecki, W.M., Schaul, T., Leibo, J.Z., Silver, D., Kavukcuoglu, K., 2017. Reinforcement learning with unsupervised auxiliary tasks. International Conference on Learning Representations .

Jiménez-Raboso, J., Campoy-Nieves, A., Manjavacas-Lucas, A., Gómez-Romero, J., Molina-Solana, M., 2021. Sinergym: a building simulation and control framework for training reinforcement learning agents, in: Proceedings of the 8th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation, ACM. pp. 319–323.

Karpatne, A., Atluri, G., Faghmous, J.H., Steinbach, M., Banerjee, A., Ganguly, A., Shekhar, S., Samatova, N., Kumar, V., 2017. Theory-guided data science: a new paradigm for scientific discovery from data. IEEE Transactions on Knowledge and Data Engineering 29, 2318–2331.

Kazmi, H., Mehmood, F., Lodeweyckx, S., Driesen, J., 2018. Gigawatt-hour scale savings on a budget of zero: deep reinforcement learning based optimal control of hot water systems. Energy 144, 159–168.

Liao, Y., Chen, H., Yu, Q., 2025. Hierarchical deep reinforcement learning for HVAC setpoint scheduling with comfort-energy decomposition. Energy and Buildings 319, 114520.

Mason, K., Grijalva, S., 2019. A review of reinforcement learning for autonomous building energy management. Computers & Electrical Engineering 78, 300–312.

Mnih, V., Kavukcuoglu, K., Silver, D., Rusu, A.A., Veness, J., Bellemare, M.G., Graves, A., Riedmiller, M., Fidjeland, A.K., Ostrovski, G., et al., 2015. Human-level control through deep reinforcement learning. Nature 518, 529–533.

Nachum, O., Gu, S., Lee, H., Levine, S., 2018. Data-efficient hierarchical reinforcement learning. Advances in Neural Information Processing Systems 31.

Pérez-Lombard, L., Ortiz, J., Pout, C., 2008. A review on buildings energy consumption information. Energy and Buildings 40, 394–398.

Picard, D., Drgoňa, J., Kvasnica, M., Helsen, L., 2017. Impact of the controller model complexity on model predictive control performance for buildings. Energy and Buildings 152, 739–751.

Raffin, A., Hill, A., Gleave, A., Kanervisto, A., Ernestus, M., Dormann, N., 2021. Stable-baselines3: reliable reinforcement learning implementations. Journal of Machine Learning Research 22, 1–8.

Raissi, M., Perdikaris, P., Karniadakis, G.E., 2019. Physics-informed neural networks: a deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. Journal of Computational Physics 378, 686–707.

Roijers, D.M., Vamplew, P., Whiteson, S., Dazeley, R., 2013. A survey of multi-objective sequential decision-making. Journal of Artificial Intelligence Research 48, 67–113.

Schulman, J., Wolski, F., Dhariwal, P., Radford, A., Klimov, O., 2017. Proximal policy optimization algorithms. arXiv preprint arXiv:1707.06347 .

Sun, M., Zheng, H., Xu, W., 2024. State history as memory: turning building control POMDPs into approximate MDPs. Energy and Buildings 304, 113841.

Sutton, R.S., 1991. Dyna, an integrated architecture for learning, planning, and reacting. ACM SIGART Bulletin 2, 160–163.

Wang, Y., Velswamy, K., Huang, B., 2025. An integral safety metric and physics-aware surrogate for reinforcement-learning building control. Engineering Applications of Artificial Intelligence 139, 109612.

Wei, T., Wang, Y., Zhu, Q., 2017. Deep reinforcement learning for building HVAC control, in: Proceedings of the 54th Annual Design Automation Conference, ACM. pp. 1–6.

Wetter, M., Zuo, W., Nouidui, T.S., Pang, X., 2014. Modelica buildings library. Journal of Building Performance Simulation 7, 253–270.

Willard, J., Jia, X., Xu, S., Steinbach, M., Kumar, V., 2022. Integrating scientific knowledge with machine learning for engineering and environmental systems. ACM Computing Surveys 55, 66.

Yang, R., Sun, X., Narasimhan, K., 2019. A generalized algorithm for multi-objective reinforcement learning and policy adaptation. Advances in Neural Information Processing Systems 32.

Zhang, Z., Chong, A., Pan, Y., Zhang, C., Lam, K.P., 2019. Whole building energy model for HVAC optimal control: a practical framework based on deep reinforcement learning. Energy and Buildings 199, 472–490.

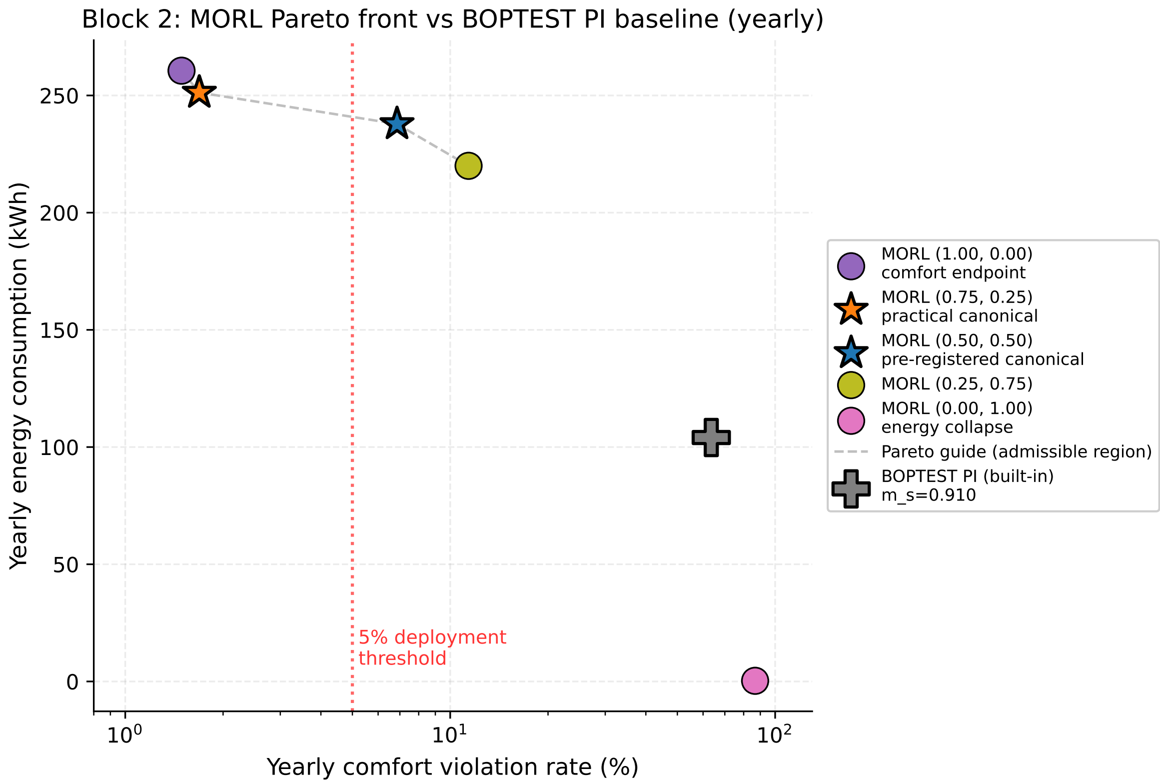

Figure 3: Block 2 control performance. Yearly Pareto scatter of the 17-D MORL controller across five preference weights, with the BOPTEST built-in PI baseline as the standard reference marker. The vertical dashed line marks the pre-registered 5% deployment-violation threshold; the gray dashed curve traces the admissible (non-collapse) region of the Pareto front. Pre-registered canonical $w = (0.50, 0.50)$ and practical-deployment canonical $w = (0.75, 0.25)$ are highlighted as starred markers and will receive error bars once seeds 43 and 44 complete (see `configs/morl_canonical_selection_log.yaml`). The $w = (0.00, 1.00)$ endpoint sits in the upper-right of the violation axis, representing the expected safety collapse at zero comfort weight.

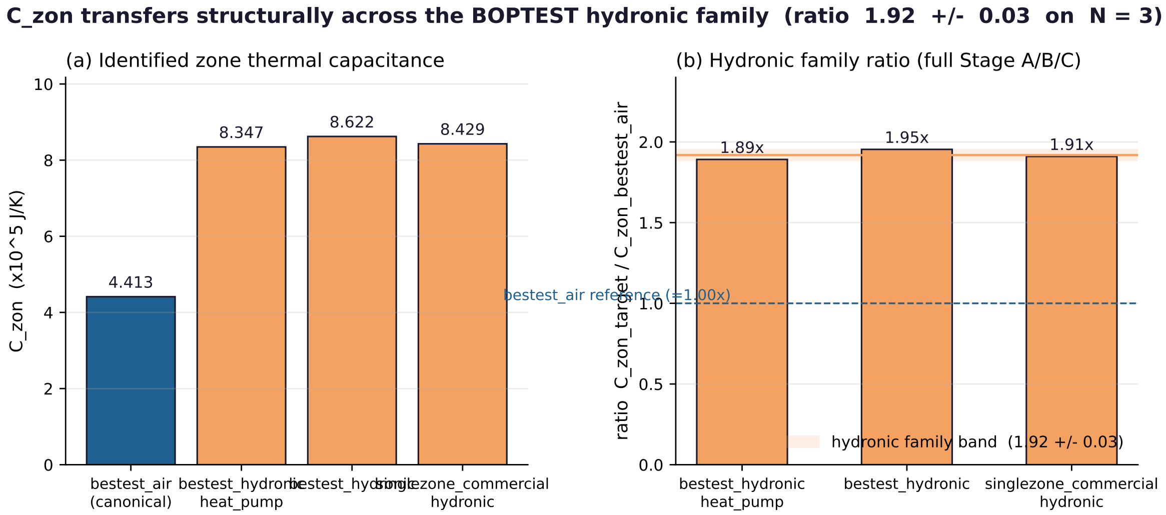

Figure 4: Identified zone thermal capacitance C_{zon} across the BOPTEST hydronic family under full Stage A/B/C recalibration. (a) Absolute C_{zon} values; (b) ratio relative to the `bestest_air` canonical. The hydronic-family band (1.91 ± 0.03) is the central numerical evidence for structural transferability of the surrogate component.

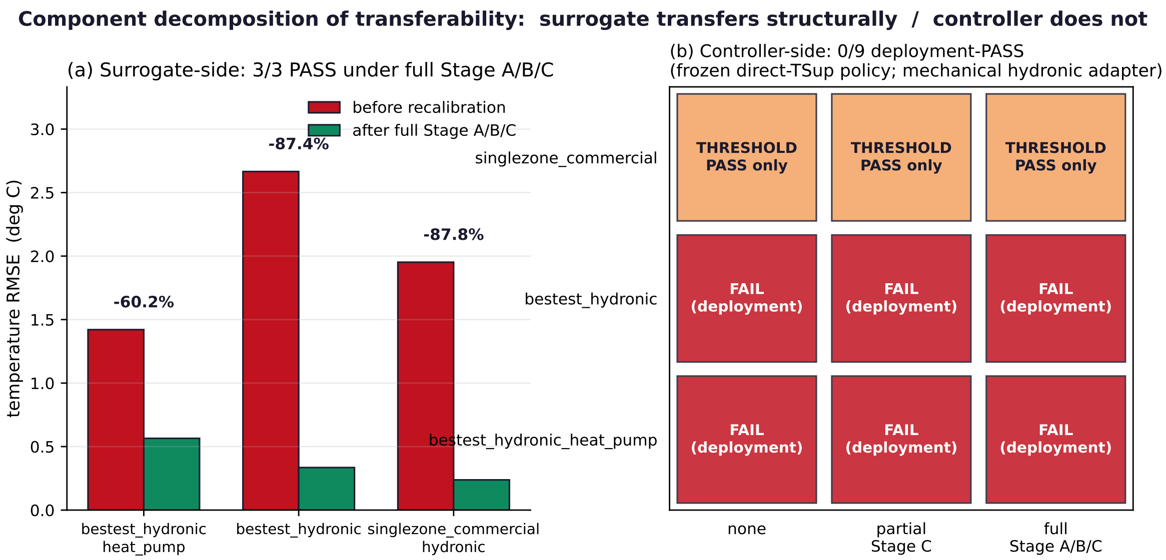

Figure 5: Component-level decomposition of transferability across the BOPTEST hydronic family ($N = 3$). (a) Surrogate-side temperature RMSE before and after full Stage A/B/C recalibration: 3/3 PASS, with 60–88% RMSE reductions. (b) Controller-side verdict matrix: 0/9 deployment-PASS under the frozen direct-supply-temperature policy routed through a mechanical hydronic adapter. The asymmetry between the two panels is the central finding of Block 3.